

THOMAS BENTIVOGLIO

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EDUCATION

Case Western University

Cleveland, OH

Bachelor of Science, Computer Science and Mechanical Engineering, Minor in Mathematics

Expected: **May 2027**

Relevant Coursework:

- Full-Stack, Data-structures, Algorithms, Discrete Math, Algorithms, Linear Algebra, Software Engineering, Programming Language Concepts, Databases
- Mechanical Manufacturing, Statics and Strength of Materials, Differential Equations, Thermodynamics, Fluid Mechanics, Design and Manufacturing, Heat Transfer

RESEARCH EXPERIENCE

Cable-Driven Hyper Redundant Robot Research (CyPhiLab)

March 2025 – Present

Undergrad Researcher

Cleveland, OH

- Designed and manufactured robots using CAD and a 3D-printer, for example a robotic arm that mimics an elephant trunk that tests different control methods and algorithms to inform future projects
- Integrated CAD models from Onshape into MuJoCo, defining tendon actuation and joint constraints, including 24 joints on a robotic arm
- Developed and tested control strategies for a cable-driven, hyper-redundant robot (“SpiRob”) using Python, ROS2, and MuJoCo for physics-based simulation, enabling a robot to lift objects up to 7 pounds
- Implemented trajectory generation and Kalman filter-based state estimation to ensure accurate robot control
- Collaborated with faculty and graduate students on robotic manipulation and human-machine interaction experiments

LEADERSHIP/ACTIVITIES

Autonomous Snowplow Club

October 2025 – Present

Club Member

Cleveland, OH

- Engineered cost-map representations for autonomous operations and obstacle avoidance
- Developed waypoint-based path planning algorithms used by a first place snowplow robot

CWRUbotix (Case Western Reserve University Robotics Club)

September 2024 – Present

Club Member

Cleveland, OH

MateROV (Submarine Robot): Mechanical and Software Team

- Designed parallel grip manipulators using SolidWorks to improve grasping power of robot
- Contributed in programming the robots’ autonomous functions using python and ROS2 in preparation for competition tasks
- Collaborated with a partner to create a pressure test chamber for an in-house-designed water-proof servo motor

Combat Robotics: Mechanical Team

- Manufactured and designed a combat robot with rapid prototyping using 3D printer and emergency carbide plating repairs to improve durability and fix damage

SKILLS/PROJECTS

- Programming: Java, Python, Java Script
 - Made the game 2048 in Java and JavaScript
 - Developed a web-app with a team that allows students to find free food
- Full Stack Development (HTML, CSS, Databases, React, MUI, etc.)
- MatLab
- SolidWorks (CSWA Certified, working towards professional)
 - Designed a water jug lift for elderly people with a team
- Certified in using lathe, mill, basics in CNC machine and water jet
 - Trained in Think[box], demonstrated skills by machining a hammer over a semester

INTERESTS: Cooking, lifting, wrestling, & old school rap music